

25 Ans: B

$$\text{Peak power} = V^2 / R \Rightarrow 2 \times 40 = (170)^2 / R$$

$$R = 361.25 \, \Omega$$

With two such light bulbs connected in series,

$$\text{the total mean power dissipated} = V^2 / 2R = (220)^2 / (2 \times 361.25) = 67.0 \, \text{W}$$